The $\tau = 0$ nearisometry question answered in the literature

A literature-status note for arXiv:math/0201098

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Status: literature already answered for the $\tau(f(X)) = 0$ branch. Vestfrid's Theorem 2(iii) proves the stronger sufficient hypothesis $\tau(f(X)) < 1/2$. This note does not claim that the survey's more ambitious $\tau(f(X)) < 1$ branch is settled.

Original question

In Section 2.12 on printed page 6 of Väisälä's survey [1], for a nonempty Q in a Banach space Y one defines

$$\rho(y, Q) = \liminf_{|t|\to\infty} \frac{\operatorname{dist}(ty, Q)}{|t|}, \qquad \tau(Q) = \sup_{y\in S_Y} \rho(y, Q).$$

The survey asks whether an ε-nearisometry $f : X \to Y$ satisfying $\tau(f(X)) = 0$, or perhaps merely $\tau(f(X)) < 1$, can be approximated by a surjective isometry.

Explicit later answer

Vestfrid's 2015 paper [2] explicitly attributes the $\tau(f) = 0$ question to Väisälä. Theorem 2(iii) states that if $f : X \to Y$ is an ε-isometry with $f(0) = 0$ and

$$\tau(f(X)) < \frac{1}{2},$$

then there is a surjective linear isometry $V : X \to Y$ such that

$$\|f(x) - Vx\| \le 2\varepsilon \qquad (x \in X).$$

The sentence immediately following Theorem 2 says explicitly that part (iii) answers Väisälä's question affirmatively, even for $\tau(f) < 1/2$. Thus the $\tau = 0$ branch is a known literature result, and the implication is stronger than the branch as asked.

Cheng, Shen, Zhang, and Zhou later give an independent exact formulation: their Theorem 2.5 makes $\tau(f) = 0$ equivalent to existence of a surjective linear isometry within 2ε [3].

Scope and provenance

This is not a new proof from the run. Vestfrid both cites the survey and says that his theorem answers its question, so the correct provenance is *literature already answered*. The theorem covers all real Banach spaces under $\tau(f) < 1/2$; it does not by itself settle the possible extension to the whole range $\tau(f) < 1$.

The original survey PDF is stored with this packet. A public author-copy full-text record for [2] was inspected, but its direct PDF endpoint returned HTTP 403 after a bounded download attempt; the exact theorem text, DOI metadata, attribution, and explicit answer sentence were nevertheless checked in the indexed full text.

References

[1] J. Väisälä, *A survey of nearisometries*, in *Papers on Analysis*, Reports of the Department of Mathematics and Statistics, University of Jyväskylä 83 (2001), 305–315; arXiv:math/0201098.

[2] I. A. Vestfrid, *Stability of almost surjective ε-isometries of Banach spaces*, J. Funct. Anal. 269 (2015), 2165–2170; DOI: 10.1016/j.jfa.2015.04.009.

[3] L. Cheng, Q. Shen, W. Zhang, and Y. Zhou, *More on stability of almost surjective ε-isometries of Banach spaces*, Sci. China Math. 60 (2017), 275–284; DOI: 10.1007/s11425-016-5155-0.